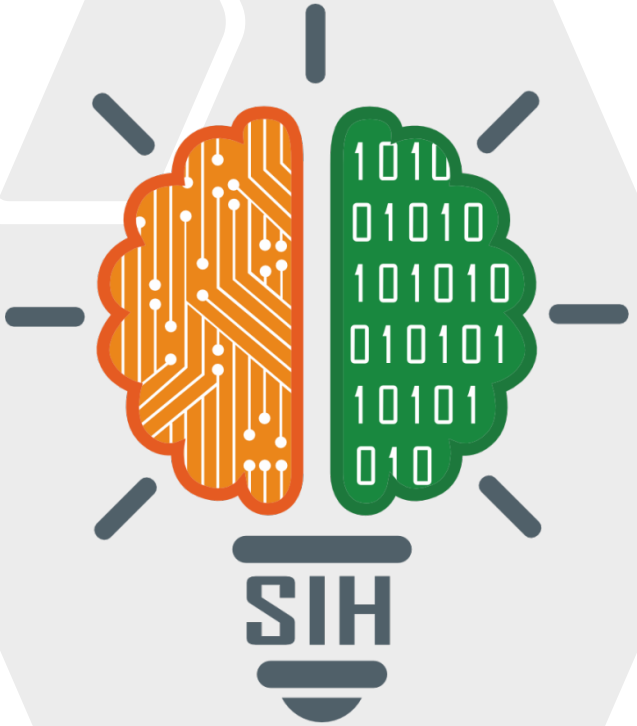
SMART INDIA HACKATHON 2026

SMART INDIA HACKATHON 2026

SatyaPatra

- **Problem Statement ID – SIH26261**
- **Problem Statement Title – AI-Powered Application Validation, Document Authentication and Regulatory Compliance Verification**
- **Theme – Smart Automation**
- **PS Category – Software**
- **Team ID – [Team ID]**
- **Team Name (Registered on portal) – [Team Name]**



[Team Name]

SatyaPatra

SMART INDIA HACKATHON 2026

Every AICTE approval document, confirmed with the authority that issued it, with the proof on screen

❖ Proposed Solution (Describe your Idea/Solution/Prototype)

Detailed explanation of the proposed solution

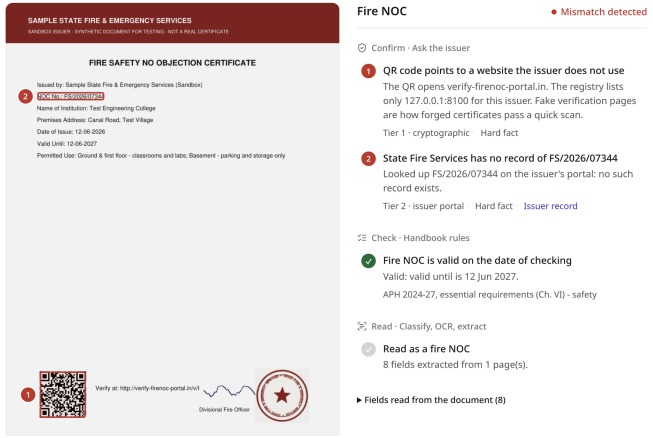
- Reads every upload: PDF text with positions, **OCR** for scans and photos
- Applies the **APH 2024-27 rulebook**: missing, expired, below-norm, mismatched
- **Asks the issuer**: signature, issuer portal, or a one-click confirm link
- Finds **tampering**: white patches, edits after signing, fake seals
- Gives **1 of the 6 PS statuses**, risk 0-100, evidence boxed on the page

How it addresses the problem

- Today a forged NOC passes until someone checks by hand: Hyderabad police booked **8 junior colleges** after **76 fire NOCs** were sent back to the issuer (Aug 2026)
- SatyaPatra makes that check the **default for every document, in under a second**
- Officers start with the **riskiest files**; every flag cites its rule and source

Innovation and uniqueness of the solution

- **Verify at source**, 3 tiers; never trusts a QR or phone printed on the document
- **Cross-institution memory**: same seal scan or faculty at two colleges
- **Satellite reality check** of built-up area
- **Seal-specimen matching** (ORB + RANSAC)
- **On-prem AI**, ₹0 licence, data stays with AICTE



Working prototype (UX4G-based console): forged NOC flagged; numbered boxes mark the evidence

120/120 forgeries caught **0/75** false alarms

0.14 s per PDF on a laptop CPU · 195 labelled test documents

1 READ
Classify each upload, OCR for scans, extract fields with their position: issuer, number, dates, address, area

›

2 CHECK
APH 2024-27 rules: missing, expired, below norm, fields that disagree across documents or with last year's record

›

3 CONFIRM
Ask the authority that issued it: digital signature / official QR, issuer portal lookup, or registered-contact confirmation

›

4 DETECT
Tampering signs: edited PDF layers, pasted text, font mismatch, identical scan pixels reused across institutions

›

5 SEE
Reality check: building footprints from open satellite data vs claimed area; lab photo GPS and capture date

›

✓ EVIDENCE REPORT
One of 6 statuses per document, proof attached, risk-ranked queue. The officer decides.

Demo video: [add unlisted link] **Live prototype:** [add link] **Code:** [add GitHub link]

[Team Name]

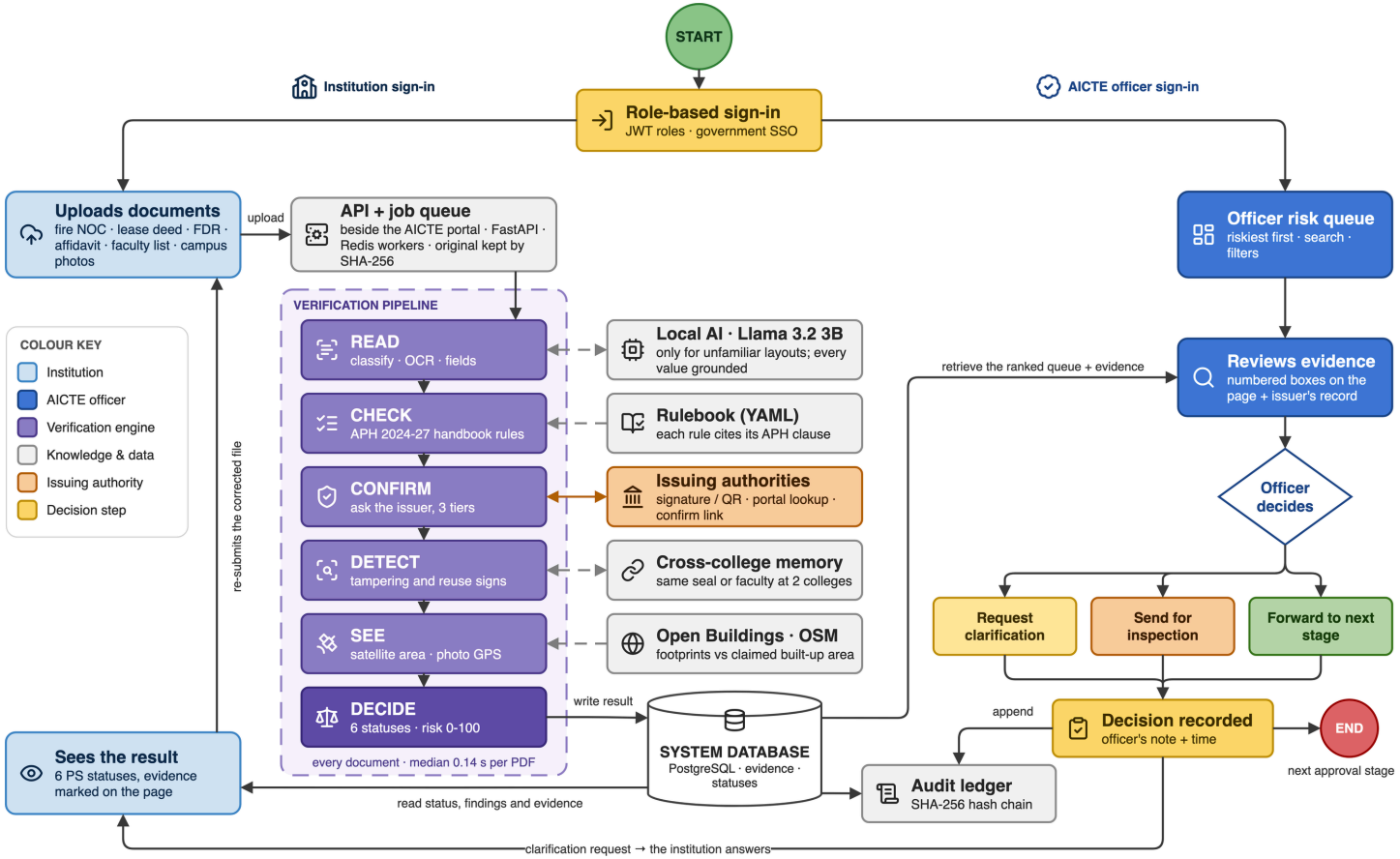
TECHNICAL APPROACH

SMART INDIA HACKATHON 2026

Technologies to be used (e.g. programming languages, frameworks, hardware)

- **FastAPI · Python 3.12**: typed async API; Python owns document AI (vs Node.js: weak PDF and vision libraries)
- **PostgreSQL 16 (+ PostGIS at scale)**: ACID evidence records; the demo runs the same code on SQLite (vs MongoDB: weak joins for audit-grade records)
- **Redis + RQ workers**: add workers in approval season (vs Celery: heavier; Kafka: overkill at this scale)
- **MinIO (S3) on NIC / MeghRaj**: originals stored by SHA-256, encrypted (vs local disk: does not scale)
- **RapidOCR (PP-OCR models, ONNX)**: CPU, layout-robust, Apache-2.0 (vs cloud OCR: data leaves AICTE)
- **pyHanko**: validates PAdES signatures to the issuer's certificate; detects edits after signing
- **OpenCV ORB + RANSAC**: rotation-proof seal matching (vs image hashes: in our tests fakes scored as close as genuine)
- **Ollama + Llama 3.2 3B**: local AI only for unfamiliar layouts, every value grounded (vs cloud LLM APIs: data leaves, per-token cost)
- **React 18 + TypeScript on UX4G 3.0 tokens**: the Government of India design system, Noto Sans, accessibility bar (vs a custom look: no GIGW alignment)
- **Role-based sign-in + SHA-256 hash chain**: government SSO via OIDC; tamper-evident log (vs blockchain: same proof, no cost)

Methodology and process for implementation (Flow Charts/Images/ working prototype)



100% open source · ₹0 licence · CPU only, no GPU

1 Built 5 engines, issuer sandbox, console, 6 passing tests

2 Measured 195 labelled documents, 8 forgery types

3 Pilot Our college's EoA, then one regional office

[Team Name]

FEASIBILITY AND VIABILITY

SMART INDIA HACKATHON 2026

Analysis of the feasibility of the idea

Technical

Built and running today; every part is mature open source

Economic

₹0 licence, no per-document fee, runs on existing NIC cloud

Operational

Plugs in beside the AICTE portal; officers keep the decision

Legal & data

On-prem, role-based, salted ID hashes; built for DPDP Act 2023

120/120 forgeries caught

0/75 false alarms

0.14 s per PDF, laptop CPU

6 of 8 forgery types hit by 2+ independent signals

Measured on our labelled test set (195 synthetic documents, forgery types modelled on reported cases). The pilot measures real-world rates.

CONFIRM: ask the authority that issued it

TIER 1 instant — **Cryptographic proof**
Digital signature traced to the issuer's certificate (India's CCA root in production) · DigiLocker-issued file · QR code only if it opens an allow-listed government domain

TIER 2 seconds — **Issuer's own portal**
Look up the NOC / FDR / licence number on the issuer's official site and compare name, date, premises

TIER 3 hours–days — **Registered-contact confirmation**
One-click "Yes, we issued this / No" request to the issuer's official contact stored in the registry. Unanswered past the SLA, the case goes to manual review

Never trust a QR code, phone number or email printed on the uploaded document itself (this is how the 2026 Delhi University fakes worked)

EVERY DOCUMENT ENDS IN ONE OF SIX STATUSES (WORDING FROM THE PS)

VERIFIED · MISMATCH DETECTED · EXPIRED · VERIFICATION PENDING · SOURCE UNAVAILABLE · MANUAL REVIEW REQUIRED

Potential challenges and risks — Strategies for overcoming these challenges

Challenge or risk	Strategy
Genuine document flagged by mistake	Soft signals alone never mark a forgery; evidence shown; institution can reply
Issuer has no API (a city fire office)	3-tier ladder; no reply within SLA goes to manual review
Phone scans and photographed documents	OCR built in (0/15 false alarms on phone scans); unclear or handwritten fields go to manual review
Forgers adapt to the detector	Issuer confirmation cannot be faked; detector is one signal of many
Approval-season peak load	Queue + more workers; issuer answers cached; ~0.14 s per PDF
Handbook changes every cycle	Rules in YAML with clause IDs, versioned by year, no code change

Viability

- **Pilot**: our college's EoA (with permission), then one AICTE regional office, then national
- **Reuse**: UGC, NCTE, PCI and state boards face the same forged-certificate problem
- **Sustainable**: open source under AICTE; no vendor lock-in
- **Scale estimate**: if ~9,000 institutions send ~30 documents each, ~2.7 lakh documents at ~1 s each take under 2 days on 32 CPU cores

[Team Name]

IMPACT AND BENEFITS

SMART INDIA HACKATHON 2026

Potential impact on the target audience

S — Students and parents
Real labs, real faculty and fire-safe buildings behind every approval

A — AICTE officers
Start with the riskiest files; routine checks automated; decisions backed by evidence

I — Honest institutions
Self-check before submitting; fewer deficiency rounds; faster approval

R — Issuing authorities
Forged copies of their certificates caught early; one-click confirmation

Benefits of the solution (social, economic, environmental, etc.)

Social

- Fire, building and lab claims actually checked
- Students get the faculty they pay for
- Approved degrees stay trustworthy

Economic

- Officer hours saved on routine checks
- Public money not lost to fake institutions or fake FDRs
- ₹0 licence, no per-document fee

Environmental

- Paperless verification end to end
- Fewer courier and travel trips to re-verify
- Satellite screening before any site visit

Governance

- Same rules for every application
- Tamper-evident record of every decision
- Reusable by UGC, NCTE, PCI, state boards

Why it matters: reported cases

8 of 76
fire NOCs sent back for checking led to 8 Hyderabad junior colleges being booked for forgery (Siasat, Aug 2026)

1 in 3
technical institutes were caught filing false data for AICTE approval (Hindustan Times, Apr 2017)

353
full-time faculty on the payrolls of 2 to 11 engineering colleges at once (DT Next, Jul 2024)

₹144 Cr
in scholarships went to 830 fake or non-operational institutions (CBI FIR via Scroll, FY18-FY22)

[Team Name]

RESEARCH AND REFERENCES

SMART INDIA HACKATHON 2026

Details / Links of the reference and research work

How SatyaPatra compares with what exists today

	Manual scrutiny	DigiLocker	Paid KYC / forgery APIs	Generic OCR	SatyaPatra
Confirms with the issuer	Partly	Its own docs	ID docs	No	Yes
NOC, lease, FDR, affidavit	Yes	Few	No	Extracts	Yes
Evidence for every flag	Varies	n/a	Score	No	Yes
Memory across institutions	No	No	No	No	Yes
Checks buildings and photos	Visits	No	No	No	Screens
No per-document fee	Yes	Yes	Paid	Paid	Yes

What we learned from earlier SIH work

- SIH 2024: an **AI-supported AICTE approval portal** (K L University) automated the workflow. We verify what flows through it
- SIH 2022 (official winners list): **AICTE one-point student/faculty verification**; **CertiSetu** scholarship verification
- Stakeholder input: [add one line from your college's AICTE coordinator]

Research papers, standards and sources

1. PS SIH26261, AICTE (sih.gov.in)
2. AICTE Approval Process Handbook 2024-27
3. Qu et al., Robust Tampered Text Detection in Document Images (DocTamper), CVPR 2023
4. Rublee et al., ORB: an efficient alternative to SIFT or SURF, ICCV 2011, doi:10.1109/ICCV.2011.6126544
5. Fischler & Bolles, Random Sample Consensus, CACM 1981, doi:10.1145/358669.358692
6. Du et al., PP-OCR: a practical ultra lightweight OCR system, arXiv:2009.09941
7. Sirko et al., Continental-scale building detection (Open Buildings), arXiv:2107.12283
8. ETSI EN 319 142 (PAdES): PDF digital signatures
9. GIGW 3.0: Indian government web guidelines (WCAG 2.1 AA)
10. Siasat, Eight colleges booked for forging HYDRAA chief's signature, Aug 2026
11. ThePrint/PTI, DU plans AI tool after manipulated documents, 20 Jul 2026
12. Careers360/PTI, Delhi basement shown as store room in documents, Jul 2024
13. Hindustan Times, One in 3 tech institutes filed false data for AICTE nod, 13 Apr 2017
14. DT Next, Ghost faculties: Anna Univ sends notices to 100 colleges, 2024
15. Scroll, 830 fake institutes got Rs 144 crore in scholarships (CBI), Aug 2023

Data and test assets (all free)

- **Labelled test set we built**: 195 documents, 8 forgery types (PS allows synthetic data)
- **Rules**: APH 2024-27; **public EoA letters** (mandatory disclosure) for real layouts
- **DocTamper** for tamper-model training; **Open Buildings + OpenStreetMap** for footprints
- **Sandbox issuer PKI** (our own CA) to test real PDF signatures end to end